3.2.P.8.1 Stability Summary and Conclusion [2439-exNA, Intrathecal Injection, Argonaut]

Stability studies have been initiated with the drug product 2439-exNA to support storage at the currently defined recommended storage condition, -20 °C ±5, and in accelerated conditions, 5 °C ±3 °C, as summarized in Table 1.

Table 1: Storage Conditions of 2439-exNA Drug Product on Stability

Drug Product Batch	Grade	Drug Substance Batch formulated	Container Closure information	Storage Conditions	
				Long Term Storage	Accelerated
240621005	GMP (Clinical use)	[redacted]	6R Schott adaptiQ® Type I glass vial, 5.3 mL fill volume 20mm Flurotec® Injection stopper 20mm flip-off 3766 White Matte seal	-20 ± 5°C	5±3°C

One GMP lot of 2439-exNA drug product, Batch # 240621005, has been produced and placed on a 36-month, two condition stability study. This stability study plan is provided in Table 3 and shows the various assays to be conducted over time for samples stored at either -20°C (supporting the intended drug product storage condition) or 5 ± 3ºC. The stability protocol is of sufficient duration to support the use of Lot 2044941 in the proposed clinical trial. Subsequently produced lots intended for clinic use will be studied using the protocol as shown in Table 3. Table 3 summarizes the available 2439-exNA drug product stability data available at the time of submission.

Table 2. Current 2439-exNA Drug Product Stability Data

Storage Condition	Batch# 240621005 (clinical and stability)
-20 ± 5°C	3 month
5±3°C	1 month

Table 3: Overview of GMP Stability Program for 2439-exNA Drug Product (Batch 240621005)

Test Method	Acceptance Criteria	Test Interval (month)					
		0 M	1 M	3 M	6 M	12 M	24 M
Appearance USP <790>	Colorless to slightly yellow solution,	R	A	B	B	B	B
Purity and impurities (denaturing AEX-HPLC)	Report purity Report any individual impurities NLT 0.2% by RRT	R	A	B	B	B	B
Purity and impurities (non-denaturing SEC)	Report purity Report any individual impurities NLT 0.2% by RRT	R	A	B	B	B	B
Assay (UV-Vis spectrophotometry)	9-11 mg/mL	R	A	B	B	B	B
Osmolality (USP <785>)	250-350 mOsm/kg	R	A	B	B	B	B
pH (USP <791>)	6.8-7.6	R	A	B	B	B	B
Sub visible particulate Matter (USP <788> Method 1)	≥10 µm: ≤6000 per container ≥25 µm: ≤600 per container	R	A	B	B	B	B
Bacterial endotoxins (USP <85>)	<0.5 EU/mL	R	A	NS	NS	B	B
Sterility (USP <71>)	No microbial growth	R	A	NS	NS	B	B

Abbreviations: R = release testing; NS = not scheduled
A = 5 ± 3 °C, B = -20 ± 5 °C

3.2.P.8.1.2 Summary of Stability Data

To date, there has not been any significant change or trend in any attribute tested, with all results being within acceptance criteria at the long-term storage condition (-20°C) for three months and the 5 ± 3°C condition for one month.

3.2.P.8.1.3 Stability Study Conduct, Initial Shelf Life and Extension Proposal

The storage temperature of -20°C is extremely conservative and based on platform knowledge of other oligonucleotide drug products no degradation of 2439-exNA is expected.

No shelf-life is claimed at this point in the development lifecycle of 2439-exNA drug product. The proposed clinical lot will continue to be studied as described in Table 3.

Based upon the stability data generated thus far and the concomitant stability monitoring of the clinical drug product, the sponsor recommends storage of the 2439-exNA drug product at -20°C.

Out-of-trend and out-of-specification test results at the long-term storage condition will trigger a formal investigation that will be performed according to local procedures at the and by the drug product manufacturer. Any failing test results will result in 2439-exNA drug product being withdrawn from clinical use. Trending of data for a quality attribute that will cause it to fail its acceptance criterion as shown in Table 2 will result in 2439-exNA drug product being withdrawn prior to the extrapolated failure date.